

Molly Carton, PhD

(240)380-0184 • mcarton@umd.edu •

Assistant Research Professor in Mechanical Engineering
University of Maryland • Maryland Robotics Center

Research and Employment



University of Maryland, College Park

Assistant Research Professor

2024–present

Bioinspired Advanced Manufacturing Laboratory • Research in design and fabrication of additive manufactured structures and soft robotic devices using 2-photon lithography.



Massachusetts Institute of Technology

Postdoctoral Fellow

2022–2024

MIT School of Engineering Postdoc Fellowship for Engineering Excellence
Portela Group • Design framework for compliant architected materials.
Mechanics, simulation, fabrication and testing of micro-scale architectures fabricated with 2-photon lithography.



University of Washington, Seattle

Graduate Research Assistant

2015–2022

Transformative Robotics Laboratory • Group-theory approach to design and fabrication of mechanical metamaterials for compliant robotics.
Solheim Additive Manufacturing Laboratory • Design, construction, and testing of novel modeling and 3D printing techniques, with emphasis on GPU parallel computing.



Logos Technologies

Summer 2016

Engineering Intern. Wide-area imaging software development, data analysis, and field testing.

Education



University of Washington, Seattle

2015–2022

PhD in Mechanical Engineering

Dissertation: *Design, Fabrication, and Application of Mechanical Metamaterials using Point Group Symmetries*

MS in Applied Math (Concurrent)

2017–2018



University of Maryland, College Park

2014–2015

Graduate studies in Mechanical Engineering



Princeton University

2010–2014

BA in Physics

Undergraduate Researcher. Shaevitz Laboratory

2013–2014

Undergraduate Researcher. Professor Daniel Marlow

2012

Active Funding

FDA BAA FY25C1AP20 - Contract No: 75F40126C00039

2026-2028

Machine Learning-Guided 3D Nanoprinting of a Retinal Vascular Phantom Library.

Co-PI. \$872,653.00

Awards and Honors

- MIT School of Engineering Postdoctoral Fellowship for Engineering Excellence (2022-24)
- NEW.Mech (New England Mechanics)– Best Teaser Talk (2023)
- MIT LEAPS Fellow (2024)
- ACTIVE: Faculty Development and Leadership Intensive, C.U. Boulder (2019)
- Mark and Lisa Tuttle Endowed Fellowship in Mechanical Engineering (2020)
- Steve and Lynn Pratt Fellowship (2020)
- Graduate School Top Scholar Fellowship Award – University of Washington (2015)

Journal Publications

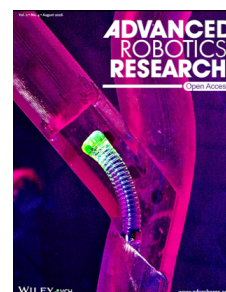
7. Felix, B. M., **Carton, M.**, Young, O., Barnes, N., Andreou, J. T., McHugh, S., McCanna, J. E., Davis, C., Reddig-Leiman, V., Kamran, A., Gandhi, D., Janowski, M., Brown, J. D., Fuge, M., Kreiger, A., Weiss, C. R., Bailey, C. R., and Sochol, R. D., “3D-Nanoprinted Fluidically Steerable Soft Robotic Microcatheters” *Advanced Robotics Research* 2.4: e202500197 (2026)

[Featured on cover].

DOI: <https://doi.org/10.1002/adrr.202500197>

Maryland Today: <https://today.umd.edu/soft-robotics-could-one-day-steer-drugs-straight-to-a-tumor>

Baltimore Sun: <https://www.baltimoresun.com/2025/10/31/soft-robot-medicine/>

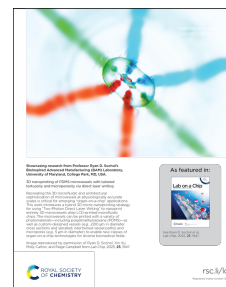


6. **Carton, M.**, Surjadi, J. U., Aymon, B. F. G, Xu, L., Portela, C. M., “Design Framework for Programmable Three-dimensional Woven Metamaterials” *Nature Communications* 17, 1581 (2026).
DOI: <https://doi.org/10.1038/s41467-026-68298-3>
MIT News: <https://news.mit.edu/2026/3d-printed-metamaterials-stretch-and-fail-design-0204>
MRS Bulletin Podcast: <https://www.mrs.org/publications-digital-content/news/mrs-bulletin-materials-news-podcast/episode/framework-designed-for-programmable-3d-woven-metamaterials>

5. Surjadi, J. U., Aymon, B. F. G, **Carton, M.**, Portela, C. M., “Double-Network-Inspired Mechanical Metamaterials” *Nature Materials*: 24, 945–954 (2025).
[Featured on cover].
DOI: <https://doi.org/10.1038/s41563-025-02219-5>
MIT News: <https://news.mit.edu/2025/mit-engineers-print-synthetic-metamaterials-strong-and-stretchy-0423>



4. **Carton, M.**, Kowalewski, J. F., Guo, J., Alpert, J. F., Garg, A., Revier, D., and Lipton, J. I., “Bridging Hard and Soft: Mechanical Metamaterials Enable Rigid Torque Transmission in Soft Robots” *Science Robotics*, 10, eads0548 (2025).
DOI: <https://doi.org/10.1126/scirobotics.ads0548>
3. Revier, D., **Carton, M.**, Lipton, J. I., “Design and Reprogrammability of Zero Modes in 2D Materials from a Single Element” *Advanced Science*, e11227 (2025).
DOI: <https://doi.org/10.1002/advs.202511227>
2. Xu, X., Qiu, Y., Chen, C., **Carton, M.**, Campbell, P.M.R., Chowdhury, A. M., Bandyopadhyay, B. C., Bentley, W.E., Smith, B. R., and Sochol, R. D., “3D Nanoprinting of PDMS Microvessels with Tailored Tortuosity and Microporosity via Direct Laser Writing” *Lab on a Chip*, 25, 1947-1958 (2025) .
[Featured on inside back cover].
DOI: <https://doi.org/10.1039/D4LC01051E>.



1. Yurtoglu, M., **Carton, M.**, and Storti, D. “Treat All Integrals as Volume Integrals: A Unified, Parallel, Grid-Based Method for Evaluation of Volume, Surface, and Path Integrals on Implicitly Defined Domains” *ASME Journal of Computing and Information Science in Engineering* 18.2 (2018): 021013.
DOI: <https://doi.org/10.1115/1.4039639>

Refereed Conference Publications

9. Campbell, P. M. R., **Carton, M.**, Dasgupta, P., Didenko, D., Dutton, T., Ghadially, A., Gomez, D., and Sochol, R. D., “Fabrication of 3D-Nanoprinted Microfluidic Transistors via Ex Situ Direct Laser Writing,” Proceedings of the 22nd Solid-State Sensors, Actuators and Microsystems Workshop (*Hilton Head 2026*), Hilton Head Island, SC, USA
8. Nishar, P., Colton, A., **Carton, M.**, Rand-Yadin, K., and Sochol, R. D., “Robust Micro-to-Macro Fluidic Interfacing of 3D-Nanoprinted Glass Microneedles via a Hybrid Direct Laser Writing Assembly Strategy,” Proceedings of the 22nd Solid-State Sensors, Actuators and Microsystems Workshop (*Hilton Head 2026*), Hilton Head Island, SC, USA

7. Lee, H-Y., Xu, X., Colton, A., **Carton, M.**, VanGessel, F. G., Saeedi, O. J., and Sochol, R. D., “Application of 3D Nanoprinting Toward a Retinal Vascular Phantom Library,” Proceedings of the 22nd Solid-State Sensors, Actuators and Microsystems Workshop (*Hilton Head 2026*), Hilton Head Island, SC, USA
[**Microsystems & Nanoengineering Springer Nature Outstanding Paper Award**]
6. Tian, Y., Xu, X., Young, O. M., Qiu, Y., Colton, A., **Carton, M.**, Smith, B. R., and Sochol, R. D., “3D Nanoprinting of High-Aspect-Ratio Porous Microvessels for Organ-on-a-Chip and Non-Animal Model Applications,” Proceedings of the 22nd Solid-State Sensors, Actuators and Microsystems Workshop (*Hilton Head 2026*), Hilton Head Island, SC, USA
5. Campbell, P., **Carton, M.**, Dasgupta, P., Didenko, D., Dutton, T., Fienkeng, C., Ghadi-ally, A., Gomez, D., Pereira, A., Walsh, B., Felix, B., and Sochol, R. D., “3D Nanoprinting Microfluidic Diodes via ex situ Direct Laser Writing,” Proceedings of the 39th IEEE International Conference on Micro Electro Mechanical Systems (*IEEE MEMS 2026*), Salzburg, Austria.
DOI: <https://doi.org/10.1109/MEMS64181.2026.11419546>
4. Tian, L., Young, O. M., **Carton, M.**, Janowski, M., Gandhi, D., Fuge, M., Kreiger, A., and Sochol, R. D., “3D-Nanoprinted Ultra-High-Aspect-Ratio Microfluidic Mixer Tubing via Two-Photon Direct Laser Writing,” Proceedings of the 29th International Conference on Miniaturized Systems for Chemistry and Life Sciences (*μ TAS 2025*), Adelaide, Australia.
3. Felix, B. M., Andreou, J. T., McHugh, S., McCann, J. E., Young, O. M., **Carton, M.**, Gandhi, D., Janowski, M., Brown, J. D., Fuge, M., Krieger, A., Weiss, C. R., Bailey, C. R., and Sochol, R. D., “Multimaterial 3D Nanoprinting of a 2 Fr Soft Robotic Microcatheter for Transarterial Chemoembolization of Liver Cancer,” In *Proceedings, 2025 IEEE-RAS Intl. Conference on Soft Robotics (RoboSoft)* Lausanne, Switzerland.
DOI: <https://doi.org/10.1109/robosoft63089.2025.11020978>
2. **Carton, M.**, Nandi, C., Anderson, A., Zhao, H., Darulova, E., Grossman, D., Lipton, J.I., Schulz, A., Tatlock, Z. “A Roadmap Towards Parallel Printing for Desktop 3D Printers.” In *Proceedings, 2021 Annual International Solid Freeform Fabrication Symposium (SFF 2021)*. Online due to COVID-19.
1. **Carton M.**, and Lipton, J. I. “Mechanical Metamaterial Continuum Materials Realized through 3D Printing using Compliant Mechanisms Techniques.” In *Proceedings, 2020 Society for the Advancement of Material and Process Engineering Conference (SAMPE 2020)*. Online due to COVID-19.

In Preparation and Review

5. Campbell, P. M. R., Chowdhury, A. M., **Carton, M.**, and Sochol, R. D., “Towards Zero-Power Fluidic Pumping via Motion-Driven 3D-Nanoprinted Microfluidic Circuitry,” Proceedings of the 30th International Conference on Miniaturized Systems for Chemistry and Life Sciences (*μ TAS 2026*), Granada, Spain. (*Accepted*)
4. Lee, H-Y., Xu, X., Colton, A., **Carton, M.**, VanGessel, F. G., Saeedi, O. J., and Sochol, R. D., “Application of 3D Nanoprinting Toward a Retinal Vascular Phantom Library,” Proceedings of the 30th International Conference on Miniaturized Systems for Chemistry and Life Sciences (*μ TAS 2026*), Granada, Spain. (*Accepted*)
3. Hill, N., **Carton, M.**, Colton, A., Liang, Y., Angueyrac, J.M. , Rand-Yadine, K., and Sochol, R.D. “3D-Printed Hydrodynamic Embryo Trapping Array for Streamlined Microinjection” (*Submitted*)
2. Young, O., Tian, L., Colton, A., **Carton, M.** Gandhi, D., Janowski, M., Brown, J.D., Fuge, M., Krieger, A., and Sochol, R.D., “Meter-Scale 3D-Nanoprinted Microfluidic Systems” (*In preparation*)
1. Campbell, P.¹, **Carton, M.**, Dasgupta, P., Didenko, D., Ghadially, A., King, R. Ryan, M. D., and Sochol, R., “Fabrication of esDLW-printed microfluidic circuitry to enhance control of desktop printed soft robotic devices” (*In preparation*)

Conference Presentations

11. Ozdil, A.G., Ma, M. Rho, Sochol, R.D., and **Carton, M.*** , “Design and Tunability of Helical Micro-actuators Fabricated with Hybrid Direct Laser Writing” Presented at ASME IDETC-CIE 2026, Houston, TX.
10. Portela, C.*, **Carton, M.**, Surjadi, J., and Aymon, B., Design and Mechanics of 3D Woven Metamaterials. 18th U.S. National Congress on Computational Mechanics (US-NCCM 2025), Chicago, IL.
9. Portela, C.*, Surjadi, J., **Carton, M.**, and Aymon, B., “3D Woven Metamaterials via Additive Manufacturing and Computational Design” Presented at Materials Research Society (MRS) 2024 Fall Meeting, Boston, MA.
8. **Carton, M.***, Surjadi, J., Aymon, B., and Portela, C., “Design Framework for Micro-printed 3D Woven Architected Materials” Presented at ASME IDETC-CIE 2024, Washington, DC.

7. **Carton, M.***, Surjadi, J., Aymon, B., and Portela, C., “Design Framework for Microscale 3D Woven Architected Materials” Presented at Engineering Mechanics Institute 2024 (EMI), Chicago, IL.
6. **Carton, M.***, Surjadi, J., Aymon, B., and Portela, C., “Design Framework for Microscale 3D Woven Architected Materials” Presented at ACM Symposium on Computational Fabrication 2023 (SCF), New York, NY.
5. **Carton, M.***, Surjadi, J., Aymon, B., and Portela, C., “Generalized Framework for Woven Architected Lattices.” Presented at NEW.Mech 2023, Boston, MA.
Best Teaser Talk.
4. **Carton M.***, and Lipton, J. I. ”Axial Point Group Auxetics with Emergent Rotational Responses.” Presented at U.S. National Congress for Theoretical and Applied Mechanics 2022 (USNC/TAM), Austin, TX.
3. **Carton, M.**, and Lipton, J. I.* “Axial Point Group Auxetics with Emergent Rotational Responses.” Presented at Materials Research Society (MRS) 2022 Spring Meeting, Honolulu, HI.
2. **Carton, M.***, and Ganter, M. “Fast and Simple Printing of Graded Auxetic Structures”. Presented at 2019 Annual International Solid Freeform Fabrication Symposium (SFF) Austin, TX, USA. *In Proceedings.*
1. Storti, D.*, Yurtoglu, M., **Carton, M.**, and Uchytel, C. “A Modern Voxel Approach to Numerical Integration on Implicit Domains using Python/Numba/CUDA” Poster presented at 2018 NVIDIA GPU Technology Conference (San Jose, CA).

Invited Talks

- UMD Prince George’s Community College Interns “Lunch and Learn” Seminar Series 2026
“Microfluidics and Microrobotics with Two-Photon Direct Laser Writing”
- University of Massachusetts, Amherst MIE Research Seminar 2026
“Pushing Boundaries of Soft Additive Manufacturing: from Design to Experiment”
- UMD Mechanical Engineering: Tubaldi Group Guest Speaker 2025
“Building Mechanical Metamaterials with Reduced Representations”

Teaching

Course Creator/Instructor

University of Maryland – Department of Mechanical Engineering

Vertically Integrated Projects (VIPs): Exoskeleton Explorers: Arthropod-Inspired
3D-Printed Robots S2025, A2025, S2026, A2026

Course Instructor

University of Maryland – Department of Mechanical Engineering

ENME 416/744: Additive Manufacturing Instructor: Spring 2026
Co-instructor: Spring & Autumn 2025

Guest Lecturer

University of Washington – Department of Mechanical Engineering

ME 498/599: Special Topics in Mechanical Engineering–Computational Design and
Additive Manufacturing

“Lattices and Groups” Guest talk, Winter 2022

“3D representations” Guest talk, Winter 2021

Graduate Teaching Assistant

University of Washington – Department of Mechanical Engineering

ME 480: Introduction to Computer-Aided Technology S2017; A2020, 2021, 2022

University of Washington – Department of Mechanical Engineering

ME 498/599: Topics in Additive Manufacturing. Autumn 2017 & 2021

Johns Hopkins Center for Talented Youth Summer Program

Principles of Engineering Design Summer 2019

University of Washington – Department of Mechanical Engineering

ME 230: Kinematics and Dynamics–*Online due to COVID-19* Spring 2020

Advising

Thesis Committee Membership

Mark Rosser M.S. Summer 2026

3-D Printed Microfluidic Capillary Circuits for the Automation of Radiopharmaceutical Synthesis

Ziteng (Jack) Wen Ph.D. Spring 2026

Novel Fabrication and Optimization Strategy of Microfluidic Drug Delivery Device via Direct Laser Writing and Machine Learning

Ryan Halli M.S. Autumn 2025

Advancing Direct Laser Writing-Based Microneedle Arrays

Nathan Hill M.S. Summer 2025

3D-Printed Microfluidics for Hydrodynamic Particle Arraying

VIPs Undergraduate Research Team

Graduate Student Teaching Assistant: M. Rho Ma (Ph.D. Candidate)

Nana Metchurchlishvili	Georgia Hammond	Andrea Murillo
Jixiang (Timothy) Suhe	Nicholas Seecharan	Katherine Rynes
Daniel Weil	Nurahmed (Nur) Multezem	Jacob Thomas
Ewan An	Elianna Nahandast	Gedalya Gordon
Fedorina Daniel	Nell Yonkos	Afsah Kamran
Olivia Shi	Adele Gamage	Natalie Westerman
Isaiah Smith	William Holmes	Roni Magidson
Eleni Green	Julian Little	

Service and Outreach

UMD Mechanical Engineering Capstone Student Interview	
Subject Matter Expert: Soft Robotics	Autumn 2026
WIE RISE (Women In Engineering) Summer Research Program	
Invited Panelist	Summer 2026
MITES (MIT Introduction to Technology, Engineering, and Science)	
Student Interview (Metamaterials)	Summer 2026
UMD SPARC (Student Preparation to Advance Robotics in College) Summer Program	
Invited Guest Lecturer	Summer 2026
UMD Maryland Day 2025	
Campus-wide public family fair	Spring 2025
VIPs Team event: spider-inspired robots team demonstration and research poster	
2025 ACM Symposium on Computational Fabrication (SCF)	
Session chair—"Small Things"	Winter 2025
ENME 437/697: Data Science in Manufacturing Quality Control	
Guest Judge: Final Project Industry/Academic Review Panel	Spring 2025
ENEE 605: Design and Fabrication of Micro-Electro-Mechanical Systems (MEMS)	
Guest Judge: Final Project Competition	Winter 2024
MIT Mechanical Engineering Research Exhibition (MERE)	
Research Poster Contest Judge	Spring 2023
UW PhD student advisory on UW Tacoma Mechanical Engineering ABET accreditation (2019; initial accreditation achieved 2022).	

Reviewing Activities

Nature Communications

IEEE Journal of Microelectromechanical Systems (MEMS)

Scientific Reports

ASME International Design Engineering Technical Conferences & Computers and
Information in Engineering Conference (IDETC-CIE)

International Solid Freeform Fabrication Symposium (SFF)

Professional Associations

American Society of Mechanical Engineers (ASME)

Association for Computing Machinery (ACM)

Professional References

Ryan Sochol

Associate Professor

University of Maryland, College Park

rsochol@umd.edu

Carlos Portela

Associate Professor

Massachusetts Institute of Technology

cportela@mit.edu

Mark Fuge

Professor

ETH Zürich

mafuge@ethz.ch

Frank VanGessel

Assistant Professor

Colorado State University

f.vangessel@colostate.edu